Lab 9: Post Lab Questions

Start Assignment

- Due Tuesday by 5pm
- Points 25
- Submitting a file upload

Lab 9: Post Lab Questions

Individual assignment. You and your lab partner share a board, so the measured numbers will match. The analysis and the writing must be your own.

> Everything here comes from work you already did.
>
> The register reads, the scope capture, the three clock readings, the frame timings, the three deliberate failures and the two masks were all produced during the milestones. There is nothing new to run.
>
> If you find yourself launching a simulation, you are answering a question that was not asked.

Due: First 60 minutes of your next lab session, the same deadline as the demonstration. Late submission costs 10% per hour.

Question 1: The control plane (6 points)

a. Include your scope capture of one SPI transaction. Mark the control bit, the address field and the data field, and mark the edge on which you sampled SPI_OUT. Say which edge you chose and why the other one fails.

b. Reading 232 from register 1, and reconstructing 488 from registers 1 and 2, are two different claims about your SPI master. State precisely what each one proves that the other does not.

c. Report the temperature registers at the start of your session and after the sensor had been running for several minutes. What does the change tell you, and what would a value that never moved have suggested?

Question 2: The clock, and the errata (5 points)

a. Report your three CLK_OUT readings with the register state at each. Compute the percentage error the wrong PLL band would have introduced, and state what that would have done to every number in

Milestone 3.

b. With the wrong band the sensor runs, answers SPI, and produces frames that look reasonable. Explain in two or three sentences why nothing else in this lab would have caught it — and say what that implies about trusting a device that appears to work.

> A datasheet default that is wrong for your clock is not a rare event. What is rare is having an instrument on the board that can tell you.

Question 3: Two timing regimes (6 points)

a. Report the two times from Milestone 3a and the pipelined period from 3b. Compare each with your pre lab prediction and give the percentage difference. Give the ratio between the single-frame and pipelined periods.

b. Explain what pipelining saves, and state clearly what it does **not** save. Then: one of your two periods answers “how often can I have a frame” and the other answers “how soon can I have one”. Say which is which, and name an application that cares about each.

c. Report your three exposure measurements and the exposure at which the period stopped being flat. Explain physically why the period is independent of exposure below that point and proportional to it above.

Question 4: What the counters revealed (4 points)

a. A row is 324 pixels, one invalid slot, then 324 pixels. Describe exactly what a counter that assumed 648 contiguous valid pixels would report, and say why that number is more dangerous than an error message would have been.

b. Report the three deliberate failures from Milestone 3c: which counter revealed each, and what its reading was. Rank them by how quickly you could have identified the cause from that reading alone.

Question 5: What you proved without an image (4 points)

a. Report your OR and AND masks in hex at both exposures. For each of these faults, state what the two masks would show: one data line stuck high, one stuck low, and one left unconstrained. Then say what the masks do **not** prove about the parallel bus.

b. List every property of the sensor you established in this lab, and next to each write the single number that established it.

c. FRAME_REQ is the first command in this course that returns nothing. Describe in two or three sentences how you verified something that gives no reply, and name three causes of a bad image you have already ruled out.

Submission

Upload a single PDF, `lab9_postlab.pdf`, containing your answers, the scope capture and any plots. Label the axes and give units.

Your **code goes to the Lab 9 – Milestones assignment**, uploaded before your demonstration.